From Long Term Classroom Programs and Medium / Short Classroom Program 4 in Top 10, 10 in Top 20, 43 in Top 100, 75 in Top 200, 159 in Top 500 Ranks & 3542 total selections in IIT-JEE 2012

ALL INDIA TEST SERIES

FIITJEE JEE(Advanced)-2013

ANSWERS, HINTS & SOLUTIONS

FULL TEST – II

(Paper – 2)

ANSWERS KEY			
	PHYSICS	CHEMISTRY	MATHEMATICS
Q. No.	ANSWER	ANSWER	ANSWER
1.	C	A	C
2.	C	D	C
3.	B	A	C
4.	B	B	D
5.	B	B	B
6.	D	B	B
7.	A	C	A
8.	C	A	C
9.	B	A	C
10.	A	D	B
11.	B	C	D
12.	C	A	B
1.	(A) → (r) (B) → (p) (C) → (q), (s) (D) → (q), (s)	(A) → (p, q, t) (B) → (p, r, s) (C) → (p, q, t) (D) → (q, t)	(A) → (s) (B) → (t) (C) → (p) (D)→ (q)
2.	(A) → (s) (B) → (q) (C) → (r) (D) → (q)	(A) → (q, r) (B) → (q, r) (C) → (r, s) (D) → (p, t)	(A) → (r) (B) → (r) (C) → (p) (D)→ (r)
1.	5	3	3
2.	4	7	5
3.	2	3	9
4.	4	5	0
5.	4	4	3

Physics PART – I

SECTION – A

3. $q = 2CV\left(1 - e^{-\frac{t}{2CR}}\right)$; $W_b = 2CV^2\left(1 - e^{-\frac{t}{2CR}}\right)$

4. LC = 4/200 = 0.02

5. Gravitational potential inside a shell = constant

6. In nuclear, nuclear force play major role. They are charge independent forces.

SECTION – C

1. Current is 2Ω = 0.1 A
current in circuit = 0.2 A
equivalent resistance about AB = 5Ω
e = I R = 1 volts
and also
$e = B\ \ell\ V$
$\therefore\quad V = \frac{e}{B\ell} = \frac{1}{2\times 0.1} = 5\,m/s$.

2. Force on the rod
$F - F_m = ma$
$F - \frac{B\ell}{r}(B\ell V - E) = ma$
$F + \frac{B\ell E}{r} - \frac{B^2\ell^2 V}{r} = \frac{mdV}{dt}$
solving
$F = -mV_0\omega\sin(\omega t) + \frac{B^2\ell^2}{r}V_0\cos\omega t\ - \frac{B\ell E}{r}$
Power expended by force is
$FV = -mV_0^2\omega\sin(\omega t)\cos(\omega t) + \frac{B^2\ell^2}{r}V_0^2\cos^2\omega t - \frac{B\ell E}{r}V_0\cos\omega t$
Its average over the cycle is
$\frac{B^2\ell^2V_0^2}{2r}$.

3. Let A_1 and A_2 be the cross–sectional area of the pipe at points P and Q respectively.
Let v_1 and v_2 be the velocities of oil at the points P and Q respectively.
By conservation of mass,
$Q = A_1v_1 = A_2v_2$
$\Rightarrow\quad v_2 = \left(\frac{A_1}{A_2}\right)(v_1) = 4v_1$
Applying Bernoullis's equation between points P and Q, we have

$$P_1 + \frac{1}{2}\rho v_1^2 = P_2 + \frac{1}{2}\rho v_2^2$$

$$P_2 = P_1 + \frac{1}{2}\rho(v_1^2 - v_2^2)$$

$$= P_1 + \frac{1}{2} \times 0.9[16 - 256] \times 10^3$$

$$= 280 \times 10^3 + (-108) \times 10^3$$

$P_2 = 172 \times 10^3\ Nm^{-2}$

$P_2 = 172\ (KN\ m^{-2})$.

4. $W_{net} = (2P_0v_0) - (P_0v_0) - \frac{\pi P_0 v_0}{4}$

$W_{net} = P_0v_0 - \frac{\pi P_0 v_0}{4} = (4 - \pi)\frac{P_0v_0}{4}$; Put $\pi = 3.14$

$W_{net} = \frac{0.86}{4} P_0v_0 = (0.22)(P_0v_0)$

Now,

$T_1 = \frac{P_0v_0}{R}$ $\qquad$ $\Delta U_{1\to2} = 1 \times \frac{3R}{2}[T_2 - T_1]$

$T_2 = \frac{4P_0v_0}{R}$ Thus, $\Delta U_{2\to3} = 1 \times \frac{3R}{2}[T_3 - T_2]$

$T_3 = \frac{2P_0v_0}{R}$ $\qquad$ $\Delta U_{3\to1} = 1 \times \frac{3R}{2}[T_1 - T_3]$

$\Delta Q_{1\to2} = (4.5)(P_0v_0) + (1.22)(P_0v_0) = (5.72)(P_0v_0)$

$\Delta Q_{2\to3} = -3P_0v_0 + 0 = -3(P_0v_0)$

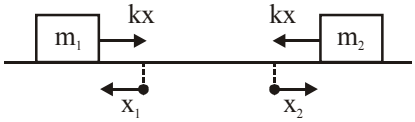
$\Delta Q_{3\to1} = -1.5\,(P_0v_0) - (P_0v_0) = -2.5(P_0v_0)$

Thus efficiency $\eta = \frac{W_{net}}{+ve\,heat}$

$\eta = \frac{0.22(P_0v_0)}{(5.72)(P_0v_0)} = 0.04$

Thus efficiency is 4%

5. The total momentum of the system in the horizontal direction is conserved. We draw the F.B.D., assuming the displacement of the block to be x_1 and x_2 in opposite directions, and the total extension x is given by,



$x = x_1 + x_2$

and $m_1x_1 = m_2x_2$

$\therefore\ m_1\frac{d^2x_1}{dt^2} = -k(x_1 + x_2)$

and $\quad m_2\frac{d^2x_2}{dt^2} = -k(x_1 + x_2)$

after suitably manipulating the equations,
we get,

$$\frac{d^2x_2}{dt^2} = -\frac{-k(m_1+m_2)}{m_1m_2}.x$$

i.e. the frequency $= \frac{1}{2\pi}\sqrt{\frac{k(m_1+m_2)}{m_1m_2}}$

$= \frac{1}{2\pi}\sqrt{\frac{300(2+3)}{2\times 3}}$

$\simeq 2.5\,\text{Hz}$.

$\therefore$ number of complete oscillations in 1 minute $= \frac{60}{2.5} = 24$.

Chemistry

PART – II

SECTION – A

1. Gem-dihalides on substitution by $\overset{\ominus}{O}H$ produces carbonyl compound, which will reacts with 2,4-DNP but 1,2-dichloroethane will give vic-diol.

2. $MnO_4^- + Fe^{+2} + 8H^{\oplus} \longrightarrow Mn^{+2} + Fe^{+3}$

   0.1 M

   0.01M 0.09 M

   $E^0_{R.P.\left(MnO_4^{-2}/Mn^{+2}\right)} > E^0_{R.P.\left(H^+/H_2\right)}$

   So permanganate electrode behave as cathode.

   E_{cell} = 1.51 – 0 = 1.51.

   Using Nernst's equation :- (n = 5)

   $$E_{MnO_4^-/Mn^{+2}} = 1.51 + \frac{0.059}{5}\log\frac{\left[MnO_4^-\right]\left[H^+\right]^8}{\left[Mn^{+2}\right]}$$

   = 1.51 + 0.11 = 1.4 V

3. Amine inversion will change the configuration of N only and not that of carbon.

5. Compound (A) will first undergo Cannizzaro reaction and the product B then will undergo esterification.

6. $\underset{\frac{500\times0.3}{760\times RT}}{NH_3} + \underset{\frac{600\times0.2}{760\times RT}}{HCl} \longrightarrow \underset{0\text{ Moles}}{NH_4Cl(s)}$ at room temperature.

   $\frac{150}{760RT}$ $\frac{120}{760RT}$

   $\frac{30}{760RT}$ 0 $\frac{120}{760RT}$

   Now total moles = $\frac{30}{760RT} + \frac{100\times1}{760RT}$

   Total P = $\frac{130\times RT}{760RT\times1.5}$ (Total volume = 1. 5 Litre)

   P in torr = $\frac{130}{1.5} = 86.66$ torr

SECTION – C

5. (S)-2,3-dimethylpentane, (R)-2,3-dimethylpentane, (S)-3-methylhexane and (R)-3-methylhexane.

Mathematics

PART – III

SECTION – A

1. $f(x)=\left(x-\frac{1}{2}\right)^3+\frac{1}{4}\left(x-\frac{1}{2}\right)+\frac{1}{2}$

$I=\int_{1/4}^{3/4} g(x)\,dx$ let $x=t+\frac{1}{2} \Rightarrow I=\int_{-1/4}^{1/4} g\left(t+\frac{1}{2}\right)dt$

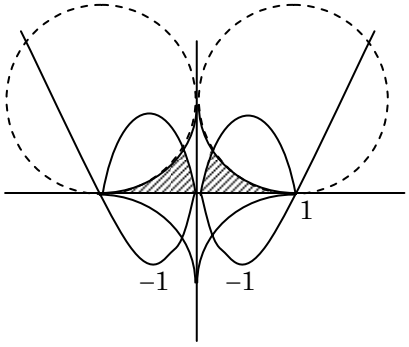
$f\left(t+\frac{1}{2}\right)=t^3+\frac{1}{4}t+\frac{1}{2} \Rightarrow f\left(f\left(t+\frac{1}{2}\right)\right)=\left(t^3+\frac{1}{4}t\right)^3+\frac{1}{4}\left(t^3+\frac{1}{4}t\right)+\frac{1}{2}$

$\Rightarrow g\left(t+\frac{1}{2}\right)=p(t)+\frac{1}{2}$ where p(–t) + p(t) = 0

$\Rightarrow I=\int_{-1/4}^{1/4}\frac{1}{2}dt=\frac{1}{4}$

2. Required area is

$2\left[\int_0^{\alpha} ex\ln x+\int_1^{\alpha}\left(1-\sqrt{1-(x-1)^2}\right)dx\right]$



3. $\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{5^n}\cdot{}^nC_r\left(\sum_{t=o}^{r-1}{}^rC_t\cdot 3^t\right)=\lim_{n\to\infty}\sum_{r=1}^{n}\frac{1}{5^n}\cdot{}^nC_r\left(4^r-3^r\right)$

$=\lim_{n\to\infty}\frac{1}{5^n}\left(\sum_{r=1}^{n}{}^nC_r4^r-\sum_{r=1}^{n}{}^nC_r3^r\right)=\lim_{n\to\infty}\frac{1}{5^n}\left(5^n-4^n\right)=1.$

4. In triangle OA_1A_2

$\cos\left(\frac{\pi}{4}\right)=\frac{2R^2-4}{2R^2}=\frac{1}{\sqrt{2}}$

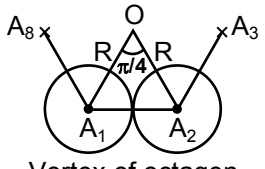
$\Rightarrow R^2=4+2\sqrt{2}$

Also

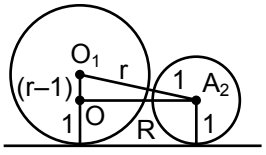
$\Rightarrow (r+1)^2-(r-1)^2=4+2\sqrt{2}$

$\Rightarrow 4r=4+2\sqrt{2}$

$\Rightarrow r=\left(1+\frac{1}{\sqrt{2}}\right)$



Vertex of octagen



5. $f\left(\frac{3\pi}{2}\right)=2\pi$

$\Rightarrow f'\left(\frac{3\pi}{2}\right)\cdot g'(2\pi)=1$

$\Rightarrow f'(x)=25(2)(2x-3\pi)^{24}+\frac{4}{3}-\sin x$

$\Rightarrow f'\left(\frac{3\pi}{2}\right)=\frac{7}{3} \Rightarrow g'(2\pi)=\frac{3}{7}$

6. Let the age of friends be a, ar, ar^2, r > 1 and money distributed is x, xr, xr^2 respectively.
$\Rightarrow (1 + r + r^2)x = (a(1 + r + r^2) + a)t$
Also after 3 years $ar^2 + 3 = 2(a + 3)$
$\Rightarrow a(r^2 – 2) = 3$
$x + 105 = (a + 3)t$ and $xr + 15 = (ar + 3)t$ (given)
$\Rightarrow (1 + r)x + 120 = (a(1 + r) + 6)t \Rightarrow 2(a + 3)t = r^2x – 120 = 2x + 210$
$\Rightarrow (r^2 – 2)x = 330$
Using $(r^2 – 2)x = 330$, $a(r^2 – 2) = 3 \Rightarrow x = 110a$
$\Rightarrow 110a + 105 = (a + 3)t \Rightarrow 5(22a + 21) = (a + 3)t$
and $110ar + 15 = (ar + 3)t \Rightarrow 5(22ar + 3) = (ar + 3)t$
$\Rightarrow \frac{22a+21}{22ar+3}=\frac{a+3}{ar+3} \Rightarrow 5ar = 7a + 6$
Where $5ar = 7a + 2(a(r^2 – 2))$ {Using $a(r^2 – 2) = 3$}
$\Rightarrow 2r^2 – 5r + 3 = 0 \Rightarrow r=\frac{3}{2}$. Hence a = 12 years

7. Given $c_n = a_1 + a_2 + a_3 + ….. + a_n$
where $a_1, a_2, ….., a_n$ are in A.P. with d = 2
and $d_n = b_1 + b_2 + b_3 + …… + b_n$
where $b_1, b_2, b_3, ….., b_n$ are in A.P. with d = 2
Also (a_n, c_n) lies on $y = px^2 + qx + r$
Now $c_n = pa_n^2+qa_n+r$ ….. (1)
$c_{n-1} = pa_{n-1}^2+qa_{n-1}+r$ ….. (2)
$\therefore$ From (1) and (2), we get
$c_n – c_{n-1} = p\left(a_n^2-a_{n-1}^2\right)+q\left(a_n-a_{n-1}\right)$
$\Rightarrow a_n=p\left(a_n+a_{n-1}\right)\left(a_n-a_{n-1}\right)+q\left(a_n-a_{n-1}\right)$
$a_n=\left(a_n-a_{n-1}\right)\left[p\left(a_n+a_{n-1}\right)+q\right]$ ….. (3)
$\left(a_n-a_{n-1}=d\right)$

10. $y = t(1 – 3t^2) \Rightarrow t = \frac{y}{x} \Rightarrow$ at A, t = – 1

Slope of tangent at A, $m = \frac{1-9t^2}{-6t}=-\frac{4}{3}$.

$\Rightarrow$ Equation of tangent at A $\Rightarrow 3y + 4x + 2 = 0$
Since tangent passes through the point $B(t_1)$ hence
$3(t_1 – 3t_1^3) + 4(1 – 3t_1^2) + 2 = 0$
$\Rightarrow (t_1 + 1)^2(3t_1 – 2) = 0 \Rightarrow t_1 = \frac{2}{3}$
$\Rightarrow B \equiv \left(-\frac{1}{3}, -\frac{2}{9}\right)$.

11. Slope of tangents at x = 0 and $m_1 = \frac{1}{\sqrt{3}}$ and $m_2 = -\frac{1}{\sqrt{3}}$

$\Rightarrow \tan\phi = \frac{\pi}{3}$.

SECTION – B

2. (A) We have $f(x) = \frac{1}{\sqrt{\ln\cos^{-1}x}}$

For domain of f(x), $\ln(\cos^{-1}x) > 0 \Rightarrow \cos^{-1}x > 1$ and $\cos^{-1}x \le \pi$

$\Rightarrow \cos\pi \le x \le \cos 1 \Rightarrow -1 \le x < \cos 1$

Hence number of integers in the domain of f(x) are 2 i.e., –1 and 0

(B) Vector normal to the plane is $\vec{n} = \hat{i} - 3\hat{j} + 2\hat{k}$ and vector along the line is $\vec{V} = 2\hat{i} + \hat{j} - 3\hat{k}$

Now $\sin\theta = \frac{\vec{x}\cdot\vec{v}}{|\vec{x}|\ |\vec{v}|} = \left|\frac{2-3-6}{\sqrt{14}\ \sqrt{14}}\right| = \left|\frac{7}{14}\right|$

Hence cosec θ = 2

(C) We have $J = \int_{-5}^{-4}(3-x^2)\tan(3-x^2)dx$. Put (x + 5) = t, we get

$J = \int_0^1(3-(t-5)^2)\tan(3-(t-5)^2)dt = \int_0^1(-22+10t-t^2)\tan(-22+10t-t^2)dt$

Now $K = \int_{-2}^{-1}(6-6x+x^2)\tan(6x-x^2-6)dx$. Put (x + 2) = z, we get

$K = \int_0^1(6-6(z-2)+(z-2)^2)\tan(6(z-2)-(z-2)^2-6)dz$

$\int_0^1(22-10z+z^2)\tan(-22+10z-z^2)dz$

Hence (J + K) = 0

SECTION – C

1. $A\left(-\frac{1}{3}, -4, 0\right)$, $B\left(0, -\frac{7}{2}, 1\right)$ and $C\left(\frac{7}{3}, 0, 8\right)$.

Hence $V = \frac{56}{18}$

2. $1+\sum_{r=1}^{10}\left(3^r\cdot{}^{10}C_r + r\cdot{}^{10}C_r\right) = 4^{10} + 10 \times 2^9$

$\Rightarrow \alpha = 1$ and $\beta = 5$

$\Rightarrow k > 4$

3. $x = {}^nC_6(3^{1/3})^{n-6}(4^{-1/3})^6$

$y = {}^nC_{n-6}(3^{1/3})^6(4^{-1/3})^{x-6}$ and $\frac{x}{y} = \frac{1}{12} \Rightarrow (12)^{n-12/3} = (12)^{-1} \Rightarrow n = 9$

4. $z_2 + z_3 = 0$

5. Let $y = \sum_{r=0}^{n} {}^{2n+1}C_r = \sum_{r=n+1}^{2n+1} {}^{2n+1}C_r$

$\Rightarrow 2y = 2^{2n+1} \Rightarrow y = 2^{2n}$

$\Rightarrow n = 3$